

PC Assembly and Disassembly Report

Task 1: Main Components Found Inside a Desktop PC

Component	Description
Motherboard	The main circuit board that connects all hardware components together.
CPU (Processor)	Executes instructions and performs calculations for the computer.
CPU Cooler	Removes heat from the processor using a heatsink and fan.
RAM (Memory)	Temporarily stores data that programs are actively using.
GPU (Graphics Card)	Processes graphics, videos, and gaming workloads.
Power Supply Unit (PSU)	Converts AC power into DC power for PC components.
SSD	Fast storage device used for operating systems and applications.
HDD	Mechanical storage device used for large-capacity data storage.
Case Fans	Improve airflow and help cool internal components.
SATA Cables	Connect storage drives to the motherboard.
Power Cables	Deliver electrical power from the PSU to components.

Task 2: Desktop PC Disassembly Process

Step 1: Shut Down and Disconnect Power

Disconnect the power cable and all external devices to prevent electrical damage.

Step 2: Remove Side Panel

Unscrew and remove the side panel of the PC cabinet to access internal components.

Step 3: Disconnect Power Supply Cables

Disconnect the 24-pin motherboard connector, CPU power cable, GPU power cable, and storage power cables.

Step 4: Remove Graphics Card (GPU)

Unscrew the GPU bracket and carefully release the PCIe slot latch before removing the graphics card.

Step 5: Remove RAM Modules

Open the retaining clips and gently pull the RAM modules out of their slots.

Step 6: Remove Storage Drives

Disconnect SATA or M.2 storage devices and remove mounting screws if necessary.

Step 7: Remove CPU Cooler

Disconnect the fan cable and unscrew the cooler from the motherboard.

Step 8: Remove CPU

Unlock the CPU socket lever and carefully lift out the processor.

Step 9: Remove Motherboard

Unscrew all motherboard mounting screws and carefully remove the motherboard from the case.

Step 10: Remove Power Supply

Unscrew the PSU from the rear of the case and remove it.

Task 3: Desktop PC Reassembly Process

Step 1: Install Power Supply

Mount the PSU first because it provides power connections needed later.

Step 2: Install Motherboard

Secure the motherboard to the case standoffs using screws.

Step 3: Install CPU

Align the CPU correctly with the socket markers and lock it in place.

Step 4: Install CPU Cooler

Attach the cooler and connect its fan cable to the CPU_FAN header.

Step 5: Install RAM

Insert RAM modules into the correct slots until the clips lock.

Step 6: Install Storage Devices

Mount SSDs/HDDs and connect SATA or M.2 interfaces.

Step 7: Install Graphics Card

Insert the GPU into the PCIe x16 slot and secure it with screws.

Step 8: Connect Power Cables

Connect motherboard, CPU, GPU, and storage power cables.

Step 9: Connect Front Panel Cables

Attach power switch, reset switch, USB, and audio connectors.

Step 10: Close Case and Test

Replace the side panel and power on the PC to verify successful assembly.

Why the Order Matters

The motherboard, CPU, and cooler must be installed before connecting other components because they form the foundation of the system. Installing parts in the proper sequence prevents cable obstruction and reduces the risk of component damage.

Task 4: Safety Precautions and Required Tools

Tools Needed

Tool	Purpose
Phillips Screwdriver	Removes and installs screws.
Anti-Static Wrist Strap	Prevents electrostatic discharge damage.
Small Container	Stores screws safely.
Flashlight	Helps see small connectors.
Cable Ties	Organizes cables after assembly.

Safety Precautions

Precaution	Importance
Disconnect Power	Prevents electric shock and component damage.
Ground Yourself	Avoids static electricity damaging hardware.
Handle Components by Edges	Protects sensitive circuitry.
Keep Workspace Clean	Prevents losing screws and parts.
Avoid Force	Prevents breaking connectors and slots.
Organize Screws	Makes reassembly easier and more accurate.

Conclusion

The desktop PC was successfully disassembled and reassembled following proper safety procedures. All major components were identified, removed, reinstalled, and connected in the correct order to ensure successful operation.